Yichen Wang, M.Sc. in Computer Science

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SUMMARY

Frontend Software Engineer with 6+ years of experience building scalable, production-grade web applications using React and TypeScript. Strong focus on performance optimization, accessibility (WCAG), test automation, and modern component architecture. Experienced in frontend migrations, API-driven UI systems, and cross-functional collaboration in Agile environments.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript (ES6+), Python, SQL
- **Frontend:** React.js, Next.js, Svelte, React Native, Tailwind CSS, Responsive Design, Accessibility
- **Backend:** Node.js (Express), Flask, RESTful APIs
- **Cloud & Infra:** AWS (EC2, S3, Lambda, DynamoDB, API Gateway), Docker
- **Databases:** PostgreSQL, DynamoDB
- **Testing:** Jest, Playwright, CI/CD (GitHub Actions)

PROFESSIONAL EXPERIENCE

IBM

Senior Software Engineer *02/2022 - PRESENT*

- Delivered 500+ frontend enhancements across three major releases of a large public-facing SPA.
- Led migration from React to Svelte, modernizing component architecture and improving bundle performance.
- Built accessible, responsive UI components aligned with WCAG standards.
- Increased test coverage from 40% to 85% using Jest and Playwright, reducing production regressions by 60%.
- Coordinated DEV → UAT → PROD releases and resolved live production issues.
- Optimized frontend performance via lazy loading, code splitting, and API optimization strategies.
- Collaborated closely with backend teams to ensure secure and reliable API integrations.

University at Buffalo

Full Stack Developer *11/2019 - 07/2021*

- Built React-based research dashboards, dynamic forms, and filtering systems.
- Improved accessibility and responsive behavior across devices.
- Developed structured UI components supporting data-heavy workflows.

INDEPENDENT PROJECT EXPERIENCE

Cross-Platform Social App (React Native + AWS + AI integrations)

- Architected and built a scalable social platform using **React/React Native, Node.js, PostgreSQL, and AWS.**
- Designed **REST APIs** supporting posts, threaded discussions, authentication, and content moderation.
- Implemented asynchronous multilingual pipelines using AWS Translate.
- Integrated AI-powered summarization and sentiment analysis via **OpenAI APIs**.
- Deployed backend services on **AWS EC2 with S3** storage and automated moderation checks (**AWS Rekognition**).

Lost & Found Pet Platform (React + Supabase + Map integrations)

- Built a geospatial React application using **Supabase (PostgreSQL + Auth)**.
- Implemented real-time updates, map-based discovery, and structured reporting workflows.
- Designed scalable data models and optimized query performance.

EDUCATION

University at Buffalo — Buffalo, NY

- M.S. in Computer Science and Engineering — May 2020
- B.S. in Computer Science, Magna Cum Laude — Dec 2018